

Patterns

25 March 2026

20:13

Q.

Q.

Q. Convert binary number to decimal.

Q. Prime factorization:

• 11001_2 $\xrightarrow{\text{multiplication}}$ $int\ count = 0;$
 $\begin{array}{r} 11001 \\ 54321 \\ \hline 0 \end{array}$ $while(num > 0);$
 $\rightarrow 1 \times 2^0 + 1 \times 2^1 + 0 \times 2^2 + 0 \times 2^3 + 1 \times 2^4 + 1 \times 2^5$
 $= 51$

$int\ count = 0;$ $int\ ans = 0;$
 $while(num > 0)\{$
 $\quad int\ multiblicen = num \% 10;$
 $\quad \text{multiblicen} \times$
 $\quad ans = ans + Math.pow(2, count);$
 $\quad count++;$

11001_2 $\xrightarrow{\text{count=0}}$ $1 \times 2^0 + 1 \times 2^1$
 $+ 0 \times 2^2$

11001_2 $\xrightarrow{\text{count=1}}$
 $\text{count} = 2$

$(base)^{power}$

$Math.pow(base, power)$
 $\downarrow$ $\downarrow$
 2 $count$
 $Count = 0$

$(2)^0, (2)^1, 2^2$

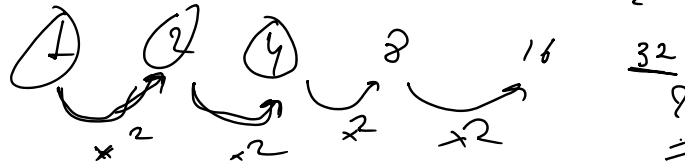
$Math.pow()$ $\rightarrow$ use $\downarrow$

Math.Pow() → vsc ↓
 → slow → algorithm ↑
 ↳ backend (log complex)

```
Scanner sc = new Scanner(System.in);
System.out.print("Enter binary number: ");
int binary = sc.nextInt();
int decimal = 0, base = 1, rem;
while (binary > 0) {
    rem = binary % 10;
    decimal = decimal + rem * base;
    base = base * 2;
    binary = binary / 10;
}
System.out.println("Decimal = " + decimal);
```

110011
 543210

$$= 1 \times 2^0 + 1 \times 2^1 + 0 \times 2^2 + 0 \times 2^3 + 1 \times 2^4 + 1 \times 2^5$$



Math.Pow();

$$2^0 = 1$$

base = 1

1 x 1 ans = 1

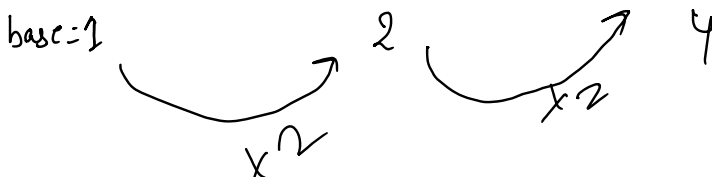
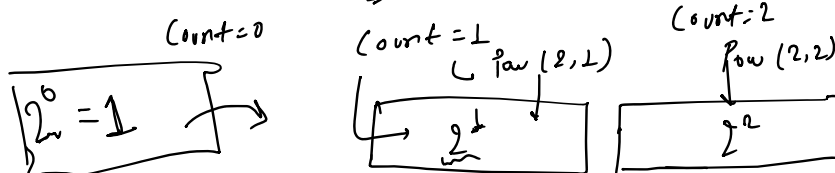
$$ans = ans + multiplier \times base$$

$$base = base \times 2;$$

$$2^3 = 2 \times 2 \times 2$$

$$1 \times 2 \times 2$$

$$= 4$$



$$2^1 \times 2 = 2^2$$

$$16 \rightarrow 1, 2, 4, 8, 16$$

$$16 \rightarrow 2 \times 2 \times 2 \times 2 = 16 \rightarrow 1 \rightarrow \text{Not a prime}$$

↳ Neither Composite

$$28 \Rightarrow 2 \times 2 \times 7 = 28$$

$$16 \rightarrow \begin{matrix} 1/6 \times & 1/5 \times & 1/4 \times & 2, 3 & 1/3 \times \\ \boxed{2 \rightarrow 16} & & & & \end{matrix}$$

num = 16

1

$$\frac{16}{2} = 8 \rightarrow \frac{8}{2} \rightarrow \frac{4}{2} \rightarrow \frac{2}{2} \rightarrow \frac{1}{2} \rightarrow X$$

$$\rightarrow 2 \ 2 \ 2 \ 2$$

$$28$$

$$\rightarrow (2 \rightarrow 28) \checkmark$$

$$\rightarrow \frac{28}{2} \rightarrow \frac{14}{2} \rightarrow \frac{7}{2} \times$$

$$2 \ 2 \ 7$$

$$\text{num} = 7$$

$$\frac{7}{3} \times$$

$$\frac{7}{4} \times$$

$$\frac{7}{5} \times$$

$$\frac{7}{6} \times$$

$$\frac{7}{7} = 1$$

$$\text{num} = 1;$$

$$\boxed{\text{num} = \text{ipl}();}$$

$$\text{int n} = \text{num};$$

$$\text{for (int } i = 2; i \leq \text{num}; i++) \{$$

$$\frac{16}{2} = 8$$

```

for (int i=2; i<=n; i++) {
    while (n % i == 0) {
        cout << i;
        n = n / i;
    }
}

```

$$\frac{16}{2} = 8$$

$$2 \cdot 2 \cdot 2$$

$$\frac{8}{2} = 4 = \frac{4}{2} = 2$$

$$2 + 1 = 3$$

$$\boxed{n=1} \quad \frac{1}{3}$$

$$i=4$$

$$2 \cdot 5^4$$

~~$$2 \cdot 64$$~~

$$127$$

$$2$$

$$\sqrt{n}$$

$$36 \rightarrow \begin{matrix} 1, 36 \\ 2, 18 \\ \vdots \\ 6, 6 \end{matrix}$$

$$\sqrt{254}$$

$$= 15$$

$$15 \cdot 9 \rightarrow 15$$

$$-\infty \text{ to } \infty$$

$$-1 \ 0 \ 1 \ 2 \ 3$$

$$2 \rightarrow 15 \quad \underline{\hspace{10em} n = \hspace{10em} 2 \ 2 \ 7 \hspace{10em}}$$

$$n=28$$

$$\sqrt{28} = 5$$

$$08$$

$$,$$

$$- \rightarrow$$

$$2 \cdot 2 \cdot 7$$

$\frac{28}{2}$
 $\frac{14}{2}$

7

(2 → 5)

$\times$ 2, $\times$ 3, $\times$ 4, $\times$ 5

2 · 2 ≠

[1 — $\sqrt{n}$]

n = 7
 Already prime

n

(1, 36)

(2, 18)

(3, 12)

(4, 9)

(6, 6)

(9, 4)

36

1 — 6

$\frac{36}{4} = 9$

$\sqrt{n} >$